

SANKALPA GIRI

+91-9777235704 sankalpagiri210@gmail.com [sankalpa-giri](https://www.linkedin.com/in/sankalpa-giri) github.com/Sankalpa-Giri

AI/ML Engineer · Backend Developer · Computer Science & Engineering (4th Year)

PROFESSIONAL SUMMARY

Undergraduate researcher and Computer Science student (**9.25 CGPA**) with publications at IEEE ICPC2T 2026 and applied research experience at DRDO's Proof & Experimental Establishment. Combines strong theoretical foundations in algorithms and machine learning with hands-on systems engineering — from NLP-driven agentic systems to offline Vision-LLM pipelines. Proficient in classical ML (supervised/unsupervised learning, model evaluation) alongside deep learning and LLM-based systems. Seeking opportunities at the intersection of research and engineering.

EDUCATION

Kalinga Institute of Industrial Technology (KIIT DU)

Bhubaneswar, India

Bachelor of Technology in Computer Science and Engineering

2023 – Present

- CGPA: **9.25 / 10.0** (as of 6th Semester)
- Relevant Coursework: Data Structures & Algorithms, Machine Learning, Operating Systems, DBMS, Computer Networks, Natural Language Processing

Range School, DRDO

Baleshwar, India

CBSE – Class XII

2022

- Aggregate: **85%**

St. Vincent Convent School

Baleshwar, India

CBSE – Class X

2020

- Aggregate: **90%**

TECHNICAL SKILLS

Programming Languages: Python, C, C++, HTML/CSS

AI / ML Frameworks: PyTorch, Scikit-learn, LangChain, Ollama, Pandas, NumPy, Matplotlib, Seaborn

ML Concepts: Supervised & Unsupervised Learning, Neural Networks, Gradient Descent, Regularisation, Model Evaluation (F1, RMSE, ROC-AUC), Cross-validation

LLMs & Vision Models: Transformers Architecture, Attention Mechanisms, Prompt Engineering, RAG Pipelines

Web & Backend: FastAPI, Flask, REST API Design, Android Studio

Databases: MySQL, ChromaDB (Vector Database)

Dev Tools: Git, GitHub, VS Code

Languages: English (Professional), Hindi (Native), Odia (Native)

INTERNSHIPS

Summer Research Intern

May 2025 – Jun 2025

Proof & Experimental Establishment (PXE), DRDO | Ministry of Defence, Govt. of India

Baleshwar, India

- Built **Project Darpan**, a prompt-driven visual intelligence system that detects objects in live camera feeds using natural language queries and Vision-LLMs (Llama 3.2 Vision).
- Architected an **offline AI inference pipeline** using LangChain + Ollama + Flask, eliminating cloud dependency and enabling real-time edge deployment in restricted defence environments.
- Integrated real-time object detection with Vision-LLM reasoning, allowing users to ask contextual questions about detected objects and scenes through plain-English prompts.
- Stack:** LangChain, Ollama, Flask, Llama 3.2, OpenCV

PROJECTS

ML Classification Pipeline – Supervised Learning on Structured Data

May 2026 – Jun 2026

- Built an **end-to-end supervised ML pipeline** on the Kaggle Titanic dataset (891 samples) – EDA, feature engineering (title extraction, family size, cabin flag), preprocessing, model training, and evaluation.
- Benchmarked Logistic Regression, Decision Tree, and Random Forest with **5-fold stratified cross-validation**; tuned Random Forest via GridSearchCV achieving **83.15% CV accuracy** and **81.6% test accuracy (ROC-AUC: 0.85, F1: 0.75)**.
- *Stack*: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Hey Shift – Android-Based NLP Voice Assistant

Jan 2026 – Apr 2026

- Engineered an end-to-end **LLM-powered voice assistant for Android** capable of understanding user intents, extracting entities, and delivering contextual, real-time responses.
- Designed a **multi-turn conversation engine** with context handling and dependency resolution, enabling coherent dialogue sessions across complex user queries.
- Integrated **Llama 3.1 (8B)** via Ollama as the local inference backend, connected to a FastAPI service layer for low-latency response delivery.
- *Stack*: LangChain, Ollama, FastAPI, Llama 3.1:8B, Android Studio

RESEARCH & PUBLICATIONS

Quantum-Optimized Ride-Pooling for Urban Mobility

IEEE ICPC2T 2026 – National Institute of Technology, Raipur

Accepted

- Co-authored a research paper proposing a **quantum computing approach to urban ride-pooling optimization**, addressing vehicle routing and passenger matching at scale.
- Formulated vehicle-passenger matching as a **combinatorial optimization problem** – algorithmically analogous to graph-based ML clustering and routing tasks.
- Collaborated with a team of 6 researchers and successfully passed the IEEE peer-review process; accepted and presented at **ICPC2T IEEE Conference 2026**.

CERTIFICATIONS

Cisco NetAcad Verified – Python Essentials 1

Issued Jun 2026

Python Programming Fundamentals | [Verify on Credly](#)

Cisco Networking Academy

- Gained a strong foundation in Python programming, including variables, data types, control structures, functions, modules, and basic problem-solving techniques.
- Developed programming skills using Python's readable syntax and industry-relevant practices, preparing for software development and automation roles.

Cisco NetAcad Verified – Python Essentials 2

Issued Jun 2026

Advanced Python Programming | [Verify on Credly](#)

Cisco Networking Academy

- Built proficiency in advanced Python concepts including object-oriented programming, exception handling, file processing, modules, packages, and code organization.
- Strengthened problem-solving and software development skills through practical programming tasks aligned with industry requirements.

SAP Certified – Backend Developer

Jun 2026 – Jun 2027

SAP Cloud Application Programming Model | [Verify on Credly](#)

Issued by SAP

- Validates expertise in leveraging the SAP Cloud Application Programming (CAP) model to design and implement extension applications for SAP S/4HANA Cloud using side-by-side extensibility on SAP BTP.

ADDITIONAL INFORMATION

Research Interests: Applied AI/ML, Large Language Models, Computer Vision, Agentic Systems, Quantum Computing

Soft Skills: Analytical Thinking, Rapid Prototyping, Cross-functional Collaboration, Technical Communication